EXP 4: Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets. (Anirudhan.S 192312056)

CODE:

```
# Import libraries from sklearn.datasets import load_iris from
sklearn.model_selection import train_test_split from
sklearn.preprocessing import StandardScaler from
sklearn.neural_network import MLPClassifier from
sklearn.metrics import accuracy_score, classification_report
# Load dataset
iris = load_iris()
X = iris.data y =
iris.target
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.2, random_state=42
)
# Normalize data scaler
= StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
# Create ANN model model
= MLPClassifier(
hidden_layer_sizes=(10,),
activation='relu',
solver='adam',
max_iter=1000,
random_state=42
)
# Train model (Backpropagation)
model.fit(X_train, y_train)
```

Predict y_pred =

model.predict(X_test)

Accuracy print("Accuracy:",

accuracy_score(y_test, y_pred))

Classification Report print("\nClassification

Report:") print(classification_report(y_test,

y_pred))

<u>OUTPUT:</u>

Accuracy: 1.0

Classification Report:

precision recall f1-score support

0 1.00 1.00 1.00 10

1 1.00 1.00 1.00 9

2 1.00 1.00 1.00 11

accuracy 1.00

30

macro avg 1.00 1.00 1.00 30

weighted avg 1.00 1.00 1.00 30

```
••• Accuracy: 1.0

    Classification Report:
                  precision    recall  f1-score   support

               0       1.00      1.00      1.00        10
               1       1.00      1.00      1.00         9
               2       1.00      1.00      1.00        11

        accuracy                           1.00        30
       macro avg       1.00      1.00      1.00        30
    weighted avg       1.00      1.00      1.00        30
```